Comprehensive Guide: AI-Powered API Test Generation with apisnap

Production-Ready Asset for Teams and Organizations

This document serves as a complete, reusable reference guide for teams and organizations to leverage **apisnap** — an AI-powered CLI tool that automatically generates API test cases. It provides comprehensive coverage of the tool's architecture, workflows, code examples, and best practices for implementing automated API testing in any project.

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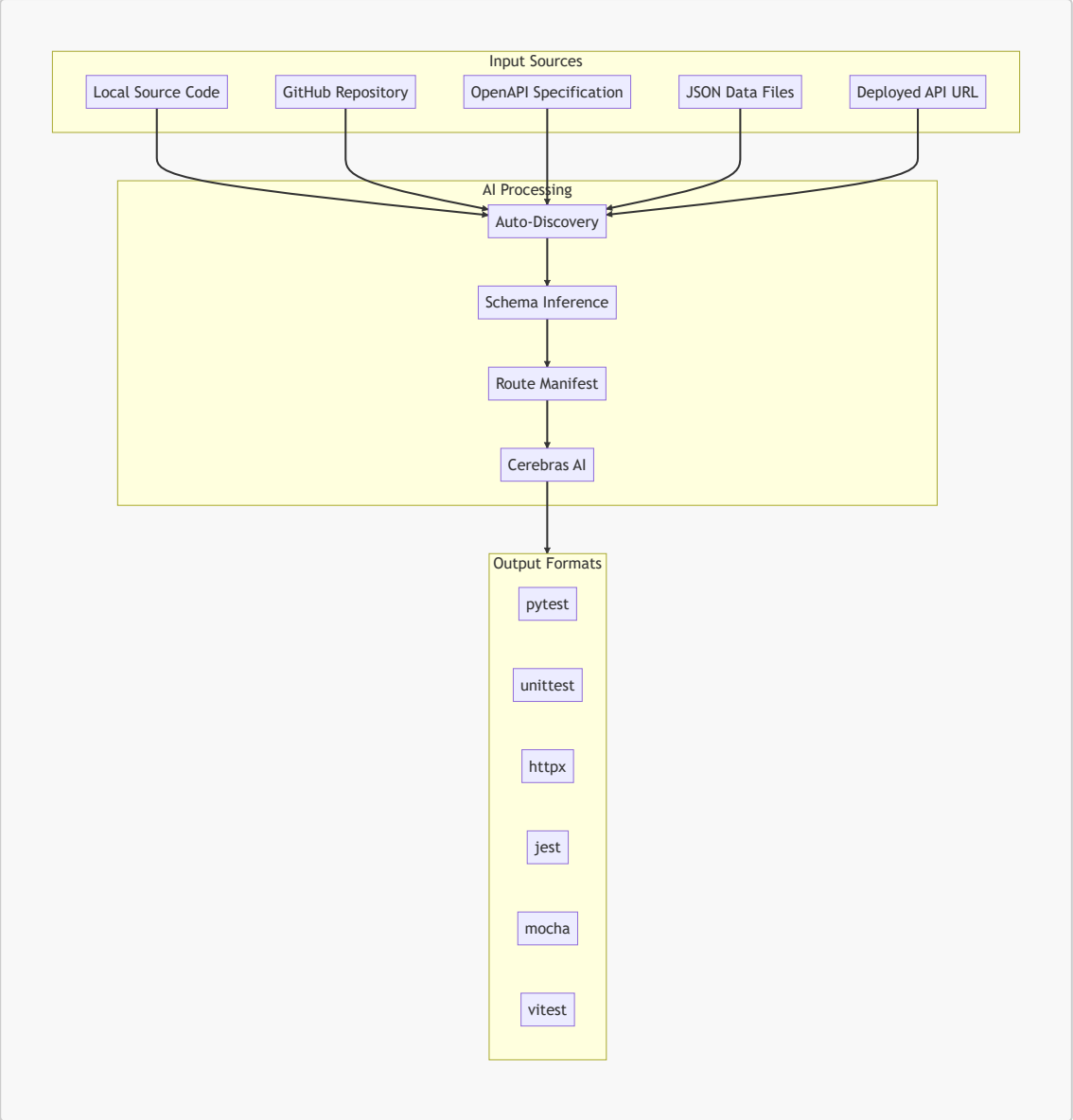
Introduction

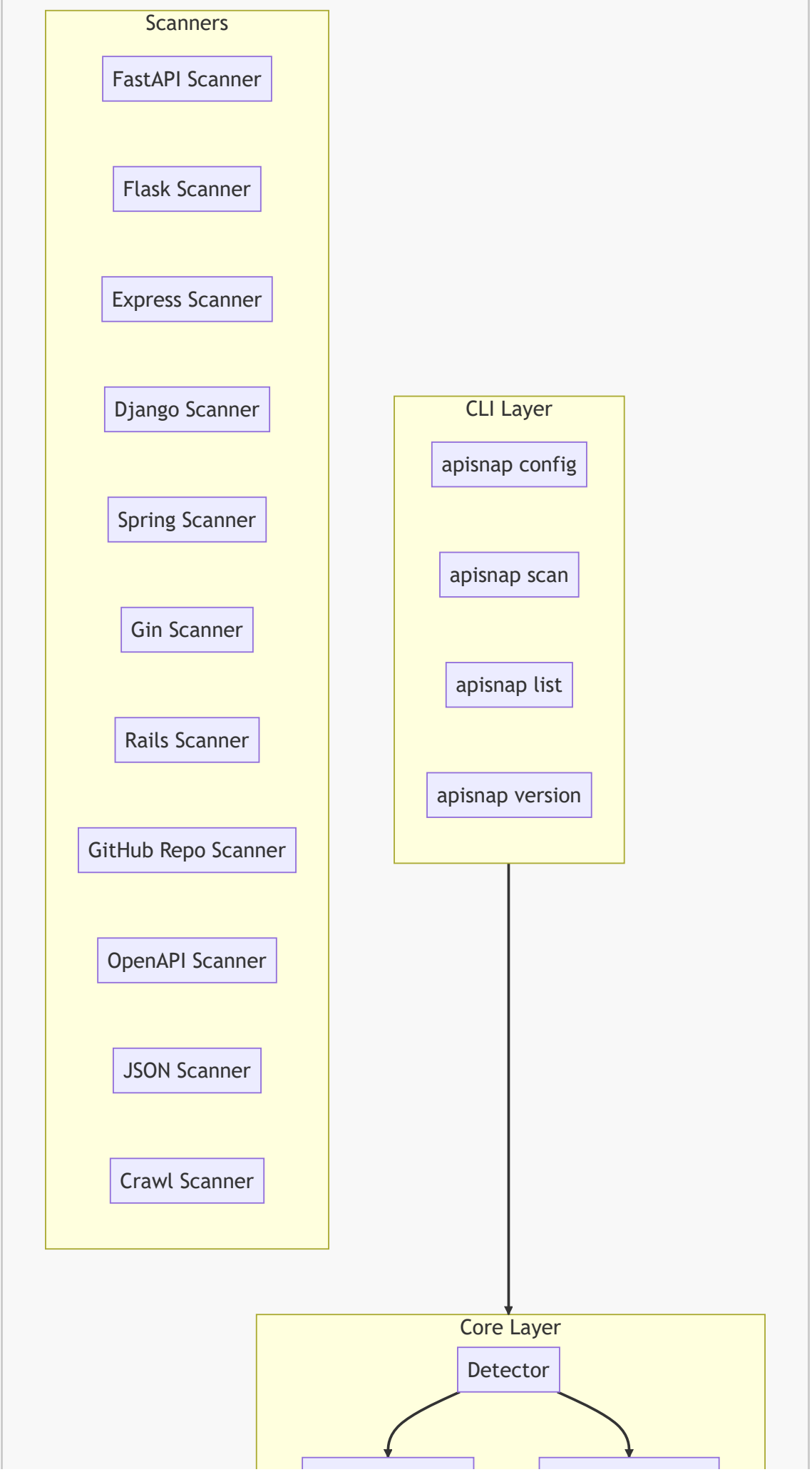
apisnap is an AI-powered CLI tool that automatically generates API test cases using Cerebras AI. It discovers API endpoints from multiple sources and produces high-quality, runnable test code in various frameworks.

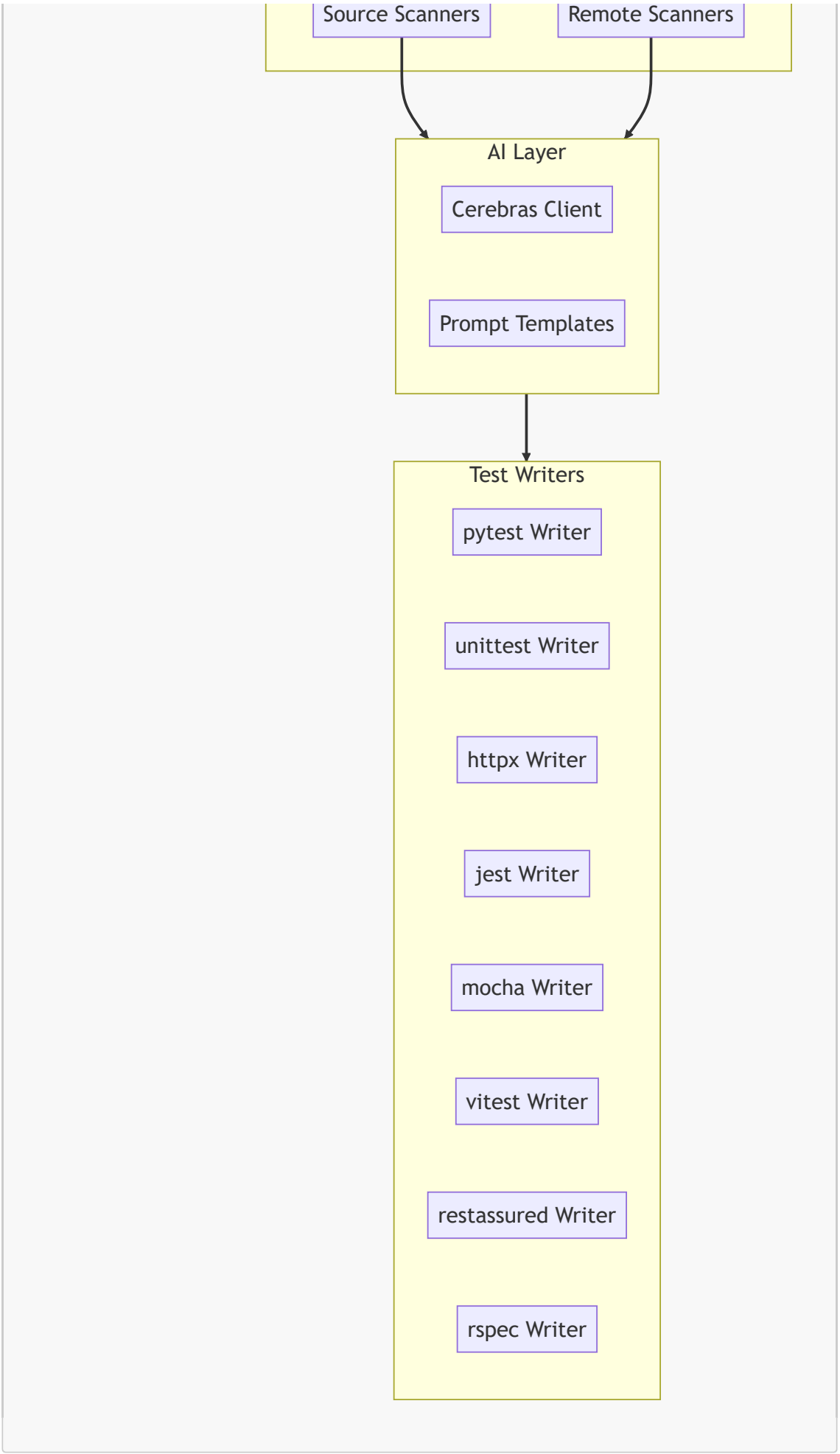
Key Capabilities



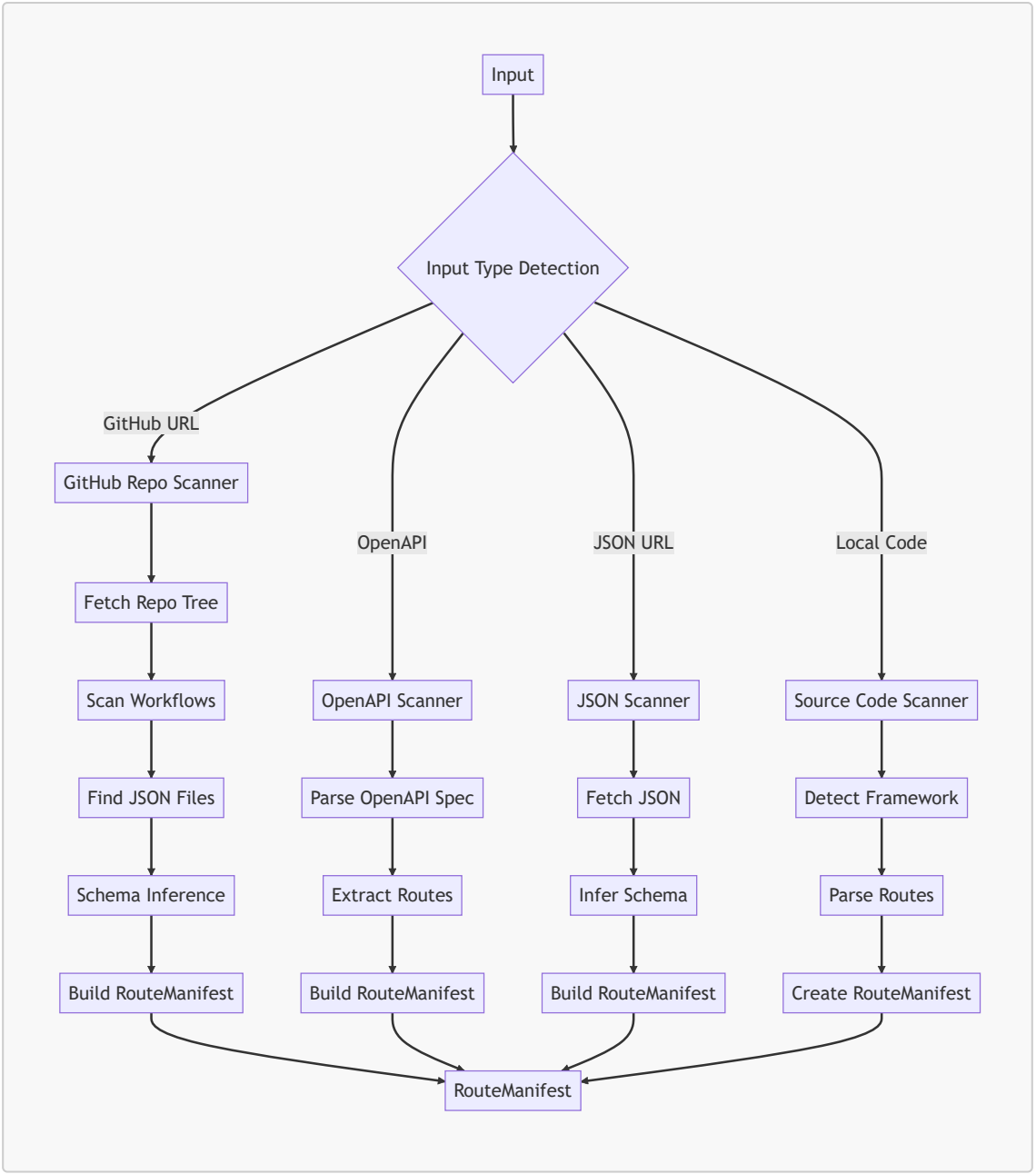
Architecture Overview

System Components


Scanners
FastAPI Scanner
Flask Scanner
Express Scanner
Django Scanner
Spring Scanner
Gin Scanner
Rails Scanner
GitHub Repo Scanner
OpenAPI Scanner
JSON Scanner
Crawl Scanner
CLI Layer
apisnap config
apisnap scan
apisnap list
apisnap version
Core Layer
Detector



Route Discovery Pipeline



Installation

Recommended: uvx (No Installation)

```
# Use directly without installation
uvx apisnap scan --url https://github.com/user/repo
```

Install with uv

```
# Add as dev dependency
uv add apisnap --dev
```

```
# Or install globally
uv pip install apisnap
```

Install with pip

```
pip install apisnap
```

Verify Installation

```
apisnap version
```

Expected Output:

```
apisnap 0.1.0
```

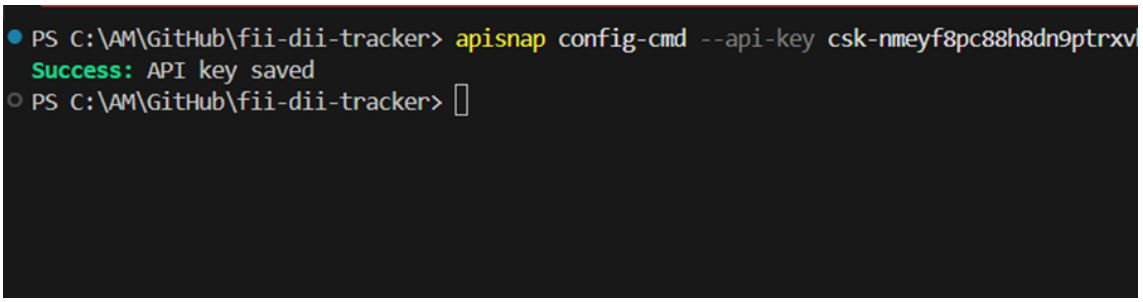
Configuration

Initial Setup

Before generating tests, you must configure the Cerebras API key.

Command: Set API Key

```
apisnap config --api-key YOUR_CEREBRAS_API_KEY
```



Alternative: Interactive Configuration

```
# Runs interactive prompt
apisnap config
```

Expected Output:

```
? Enter Cerebras API key: [hidden input]
? Default format (pytest) pytest
? Default output directory ./tests
Success: API key saved
```

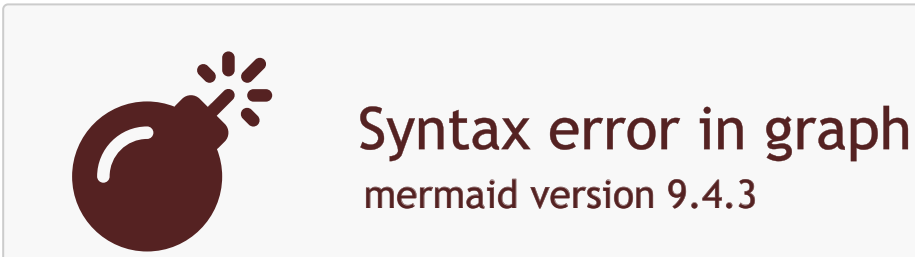
Show Current Configuration

```
apisnap config --show
```

Expected Output:

```
Cerebras API Key: csk-xxxx...xxxx (first 8 chars shown)
Default Format: pytest
Default Output: ./tests
```

Configuration File Location



Config stored at: `~/.apisnap/config.toml`

```toml
# ~/.apisnap/config.toml
[cerebras]
api_key = "csk-xxxx"           # Your Cerebras API key
model = "qwen-3-235b-a22b-instruct-2507"

[defaults]
output_dir = "./tests"
format = "pytest"
```

CLI Commands Reference

Command: scan

The primary command for scanning and generating tests.

```
apisnap scan [PATH] [OPTIONS]
```

Options

Option	Description	Example
`PATH`	Path to scan (default: current directory)	`./src`
`--url TEXT`	Remote URL (GitHub, OpenAPI, JSON)	`--url https://github.com/user/repo`
`--format TEXT`	Test framework	`--format pytest`
`--output TEXT`	Output directory	`--output ./tests`
`--framework TEXT`	Force framework detection	`--framework fastapi`
`--mode TEXT`	Force discovery mode	`--mode github`
`--dry-run`	Show routes only, no test generation	`--dry-run`
`--base-url TEXT`	Base URL for tests	`--base-url https://api.example.com`
`--verbose`	Detailed progress output	`--verbose`
`--no-ai`	Print manifest as JSON, skip AI	`--no-ai`

Command: config

Manage configuration.

```
apisnap config [OPTIONS]
```

Option	Description
`--api-key TEXT`	Set API key
`--show`	Show current config
`--format TEXT`	Set default format
`--output-dir TEXT`	Set default output dir

Command: list

List discovered routes.

```
apisnap list [PATH]
```

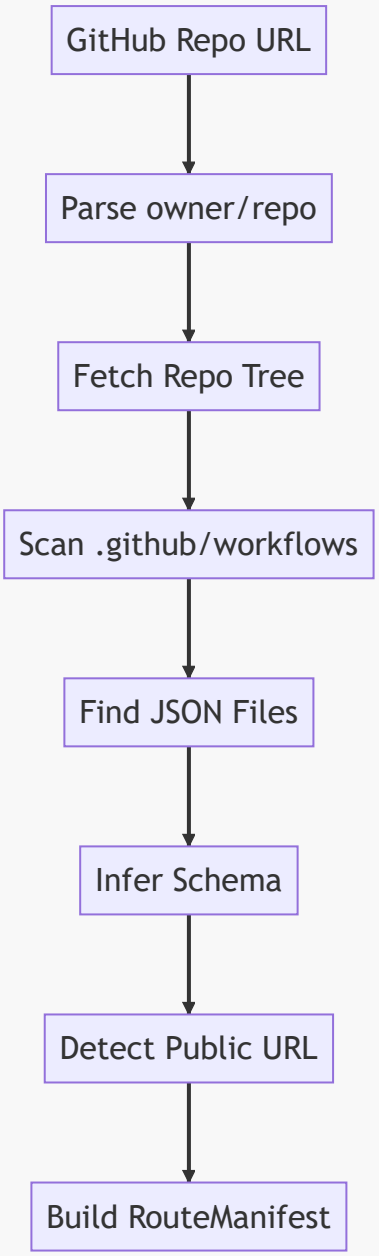
Command: version

Show version.

```
apisnap version
```

Scanning Modes

Mode 1: GitHub Repository (GitHub-as-Database)



Command

```
apisnap scan --url https://github.com/owner/repo
```

Real Example:

```
apisnap scan --url https://github.com/chirag127/fii-dii-tracker
```

```
PS C:\AM\GitHub\fii-dii-tracker> apisnap scan --url https://github.com/chirag127/fii-dii-tracker
.: Scanning for routes...

Generating Tests
================
: Generating tests...
Success: Generated 2 test files in ./tests/
PS C:\AM\GitHub\fii-dii-tracker>
```

Mode 2: Local Source Code

```
# Scan current directory
apisnap scan

# Scan specific directory
apisnap scan ./src

# Scan with framework hint
apisnap scan ./src --framework fastapi
```

Supported frameworks:

- FastAPI
- Flask
- Django REST Framework
- Express
- Spring Boot
- Gin (Go)
- Ruby on Rails

Mode 3: OpenAPI Specification

```
# From URL
apisnap scan --url https://api.example.com/openapi.json

# From file
apisnap scan ./openapi.json
```

Mode 4: JSON Data Files

```
# From URL
apisnap scan --url https://example.com/data.json

# With custom base URL
apisnap scan --url https://example.com/data.json --base-url
https://api.example.com
```

Mode 5: Dry Run (Preview Only)

```
# Preview routes without generating tests
apisnap scan --url https://github.com/user/repo --dry-run
```

Expected Output:

```
Discovered Routes
================

Route: GET /data/chords.json
  URL: https://user.github.io/repo/data/chords.json
  Method: GET
  Auth: Not Required
  Confidence: 0.95

Route: GET /data/prices.json
  URL: https://user.github.io/repo/data/prices.json
  Method: GET
  Auth: Not Required
  Confidence: 0.95
```

Code Examples

Example 1: Full Workflow

Step 1: Configure API Key

```
# Set up your Cerebras API key
$ apisnap config --api-key csk-
nmeyf8pc88h8dn9ptrxvhnr38xvrv8925ecf5t89hkn9deff

Success: API key saved
```

Step 2: Generate Tests

```
# Scan a GitHub repository
$ apisnap scan --url https://github.com/chirag127/fii-dii-tracker

.: Scanning for routes...

Generating Tests
================
```

```
⠋ Generating tests...
Success: Generated 2 test files in ./tests/
```

Step 3: Run Generated Tests

```
# Navigate to test directory
cd tests

# Run pytest
pytest -v
```

Example 2: Using Different Frameworks

Generate jest Tests

```
apisnap scan --url https://github.com/user/repo --format jest
```

Generate mocha Tests

```
apisnap scan --url https://github.com/user/repo --format mocha
```

Generate vitest Tests

```
apisnap scan --url https://github.com/user/repo --format vitest
```

Example 3: Custom Output Directory

```
apisnap scan --url https://github.com/user/repo --output ./my-tests
```

Example 4: Verbose Mode

```
apisnap scan --url https://github.com/user/repo --verbose
```

Expected Output:

```
Scanning: https://github.com/user/repo
Detected mode: github
Found 3 JSON files
Fetching: data/users.json
Fetching: data/products.json
Fetching: data/orders.json
Generating tests...
Writing: test_api_users.py
Writing: test_api_products.py
Writing: test_api_orders.py
Success: Generated 3 test files in ./tests/
```

Example 5: Show Routes Only (Dry Run)

```
apisnap scan --url https://github.com/user/repo --dry-run
```

Expected Output:

```
Discovered Routes
================

┌──────────────────────────────────────────────────────────────────────
──────────┐
│ Method │ Path                    │ URL                          │
Conf  │
├────────┼─────────────────────────┼──────────────────────────────────
──┼───────┤
│ GET    │ /data/users.json        │ https://u.gi.io/repo/u.json
│ 0.95 │
│ GET    │ /data/products.json    │ https://u.gi.io/repo/p.json    │
0.95 │
│ GET    │ /data/orders.json     │ https://u.gi.io/repo/o.json     │
0.95 │
└──────────────────────────────────────────────────────────────────────
──────────┘
```

Example 6: Custom Base URL

```
apisnap scan --url https://github.com/user/repo --base-url
https://api.example.com
```

Example 7: No AI Mode (JSON Output)

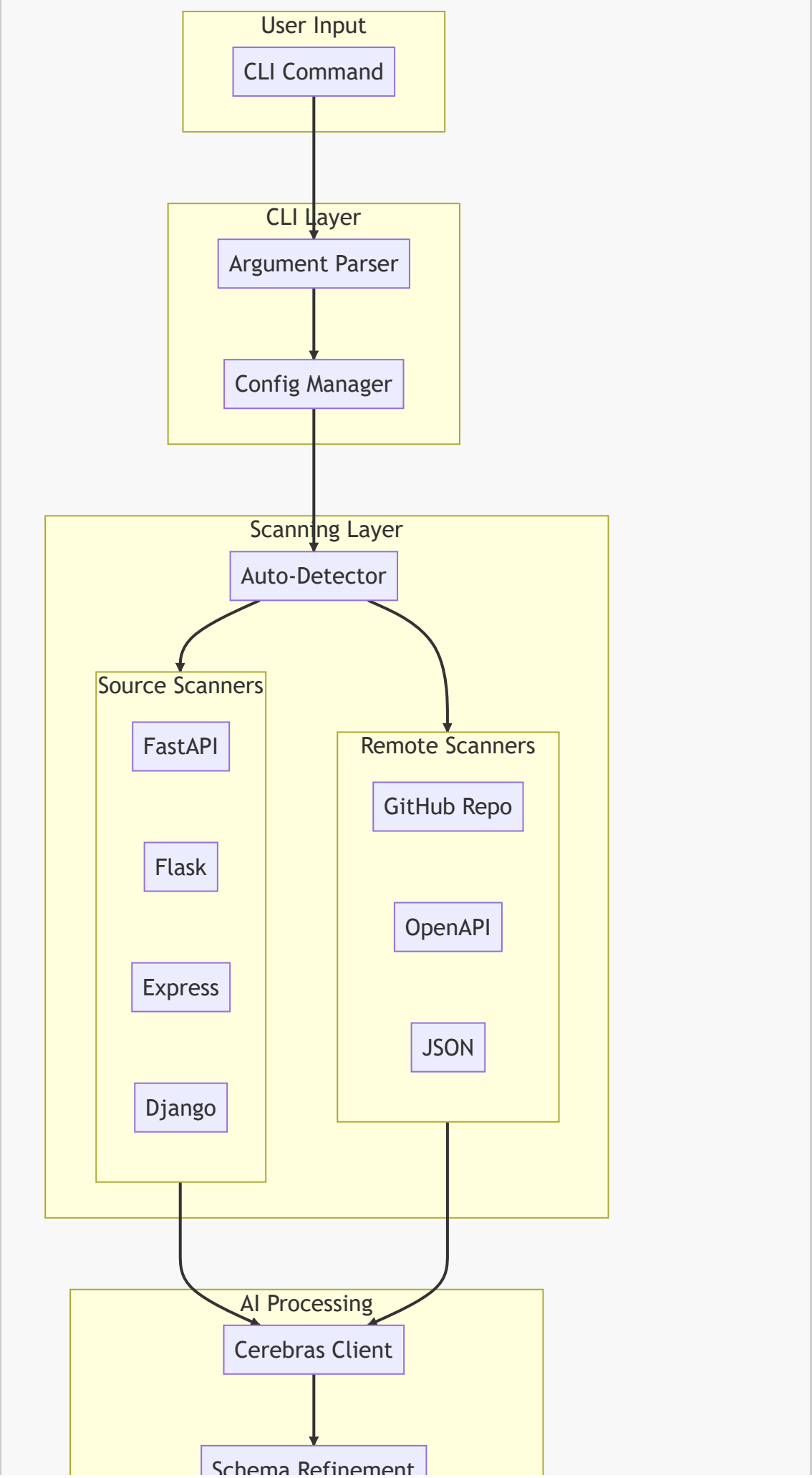
```
apisnap scan --url https://github.com/user/repo --no-ai
```

Expected Output:

```
{
  "routes": [
    {
      "method": "GET",
      "path": "/data/users.json",
      "public_url": "https://user.github.io/repo/data/users.json",
      "response_schema": {
        "type": "array",
        "items": {
          "type": "object",
          "properties": {
            "id": {"type": "integer"},
            "name": {"type": "string"}
          }
        }
      },
      "confidence": 0.95,
      "source": "github-data-repo"
    }
  ],
  "base_url": "https://user.github.io/repo",
  "framework": "github-data-repo",
  "project_name": "user/repo"
}
```

Mermaid Diagrams

Diagram 1: Complete System Architecture


User Input
CLI Command
CLI Layer
Argument Parser
Config Manager
Scanning Layer
Auto-Detector
Source Scanners
FastAPI
Flask
Express
Django
Remote Scanners
GitHub Repo
OpenAPI
JSON
AI Processing
Cerebras Client
Schema Refinement

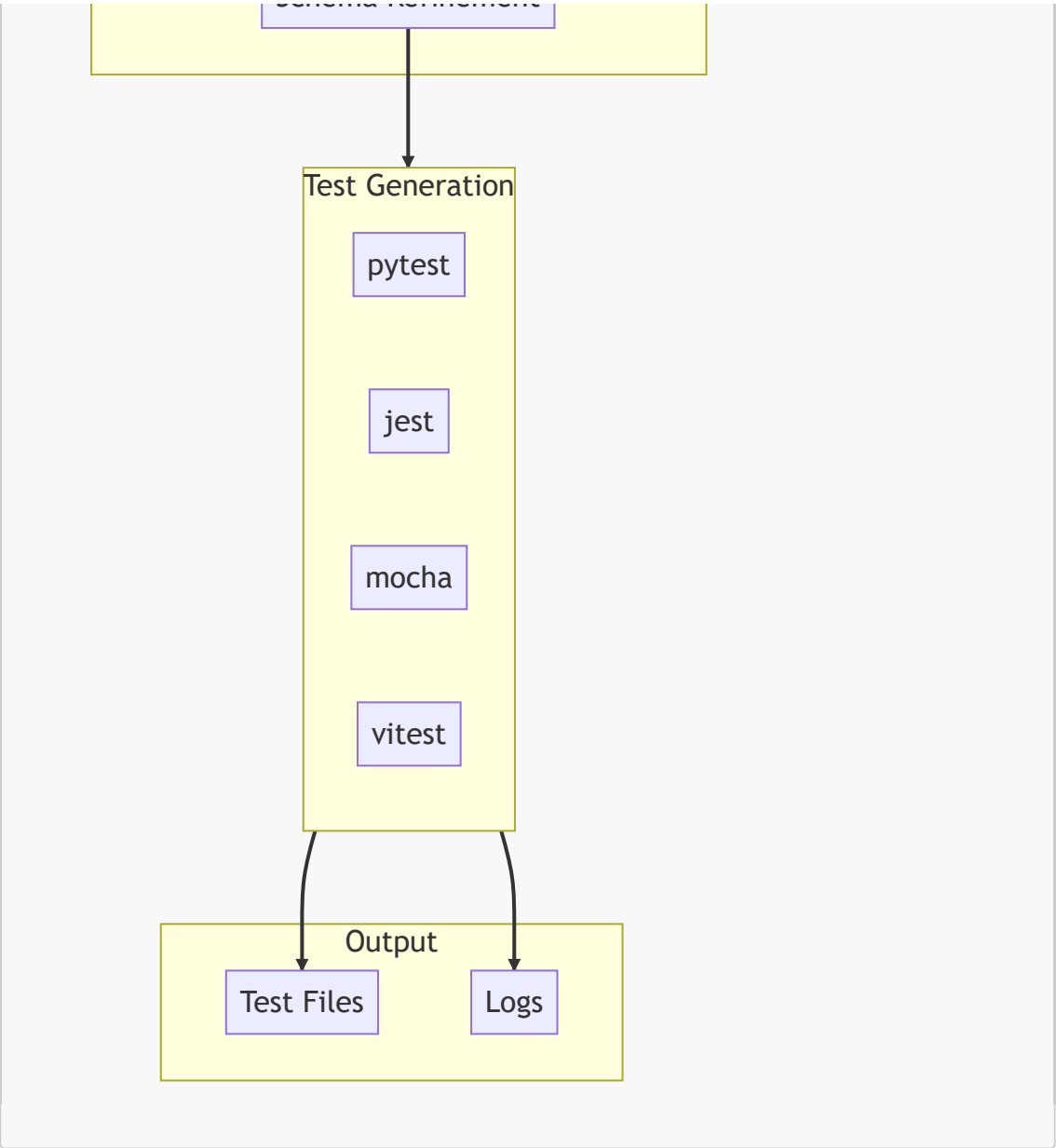


Diagram 2: GitHub-as-Database Pattern

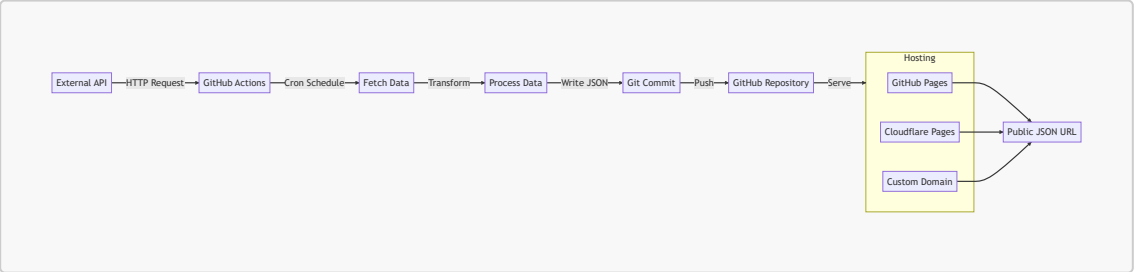


Diagram 3: Route Discovery Flow

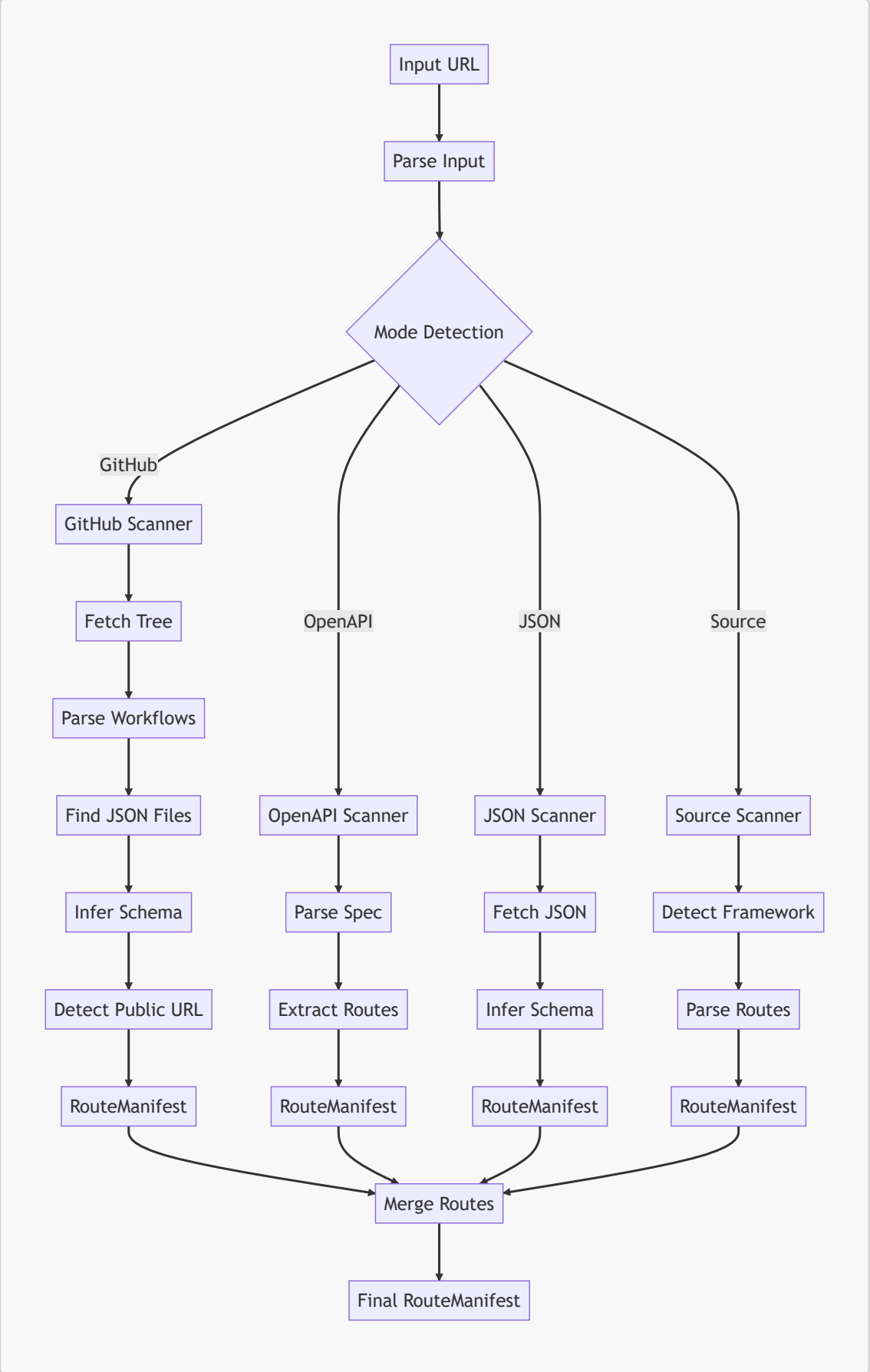
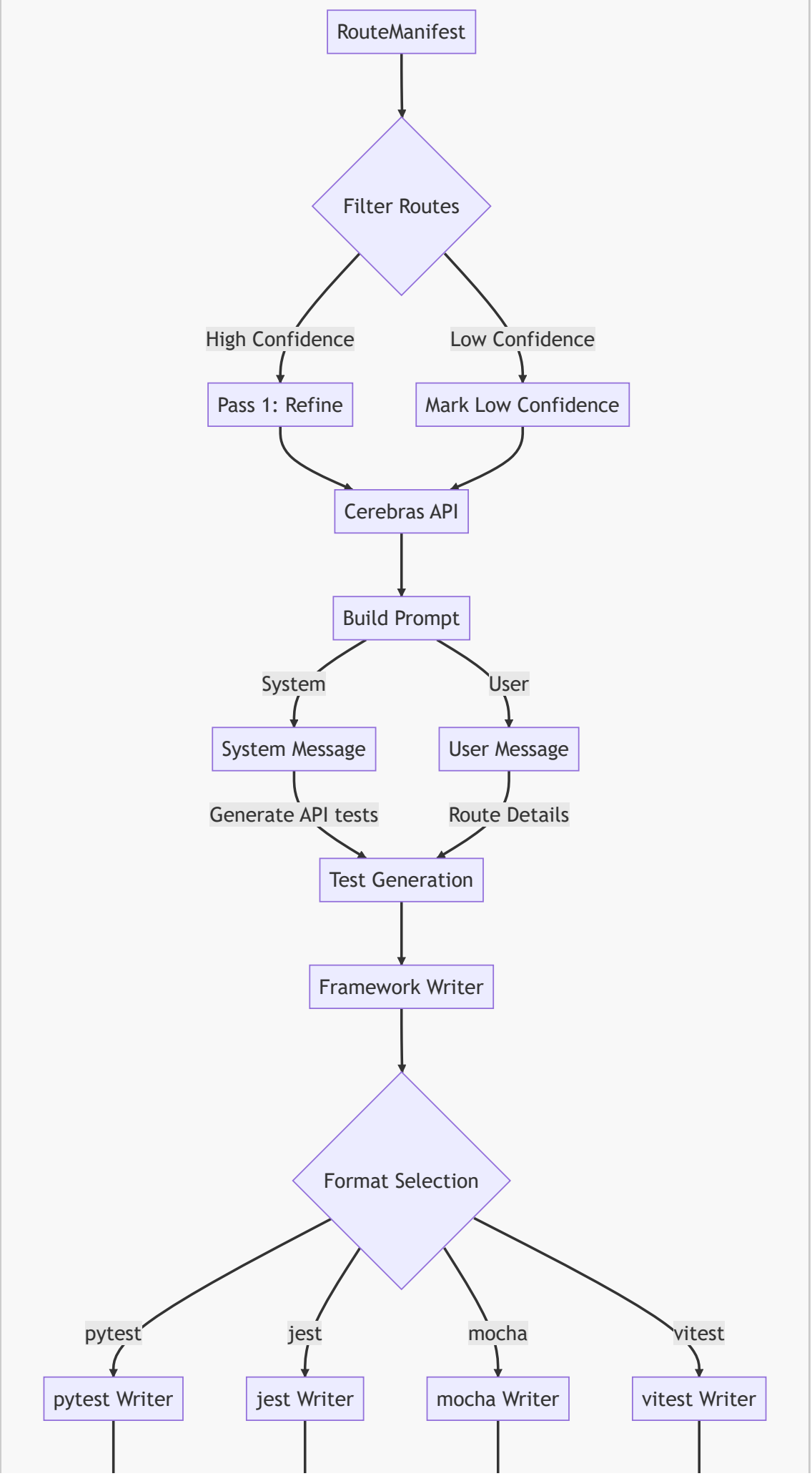


Diagram 4: AI Test Generation Pipeline


RouteManifest
Filter Routes
High Confidence
Low Confidence
Pass 1: Refine
Mark Low Confidence
Cerebras API
Build Prompt
System
User
System Message
User Message
Generate API tests
Route Details
Test Generation
Framework Writer
Format Selection
pytest
jest
mocha
vitest
pytest Writer
jest Writer
mocha Writer
vitest Writer

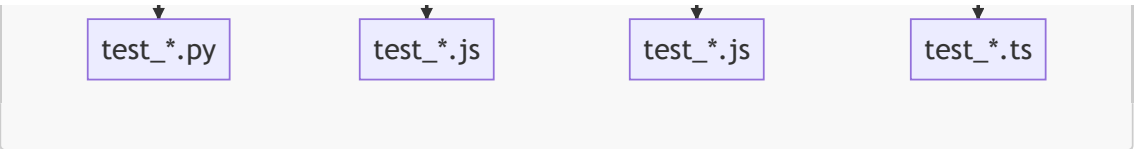


Diagram 5: Test Categories Generated

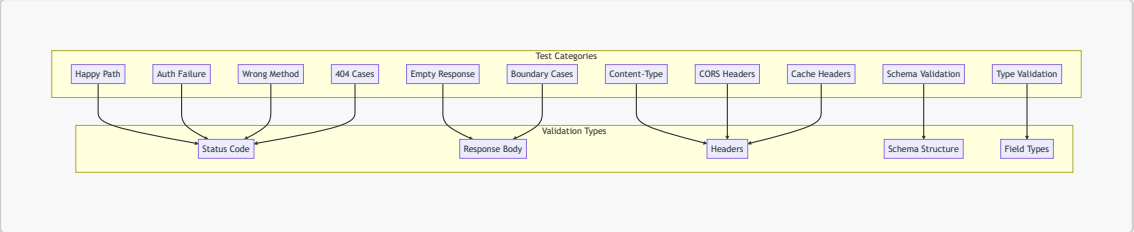


Diagram 6: Data Flow Architecture

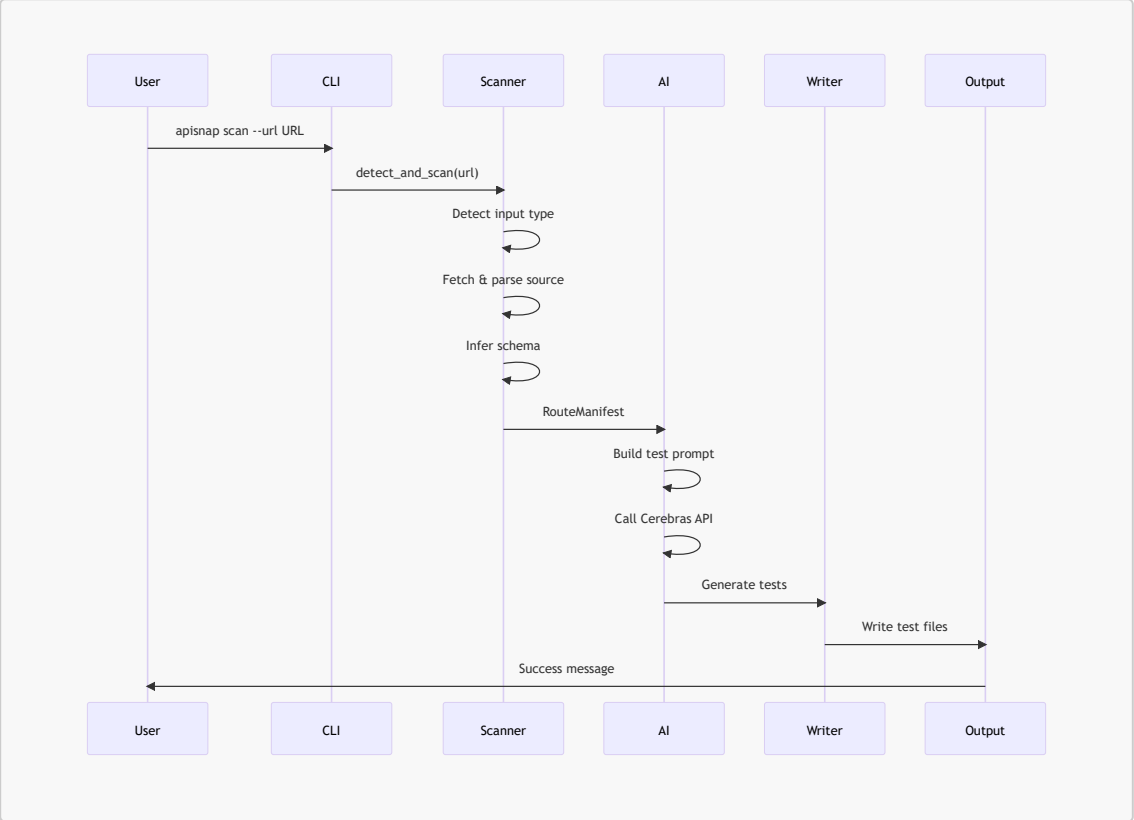


Diagram 7: Configuration Flow

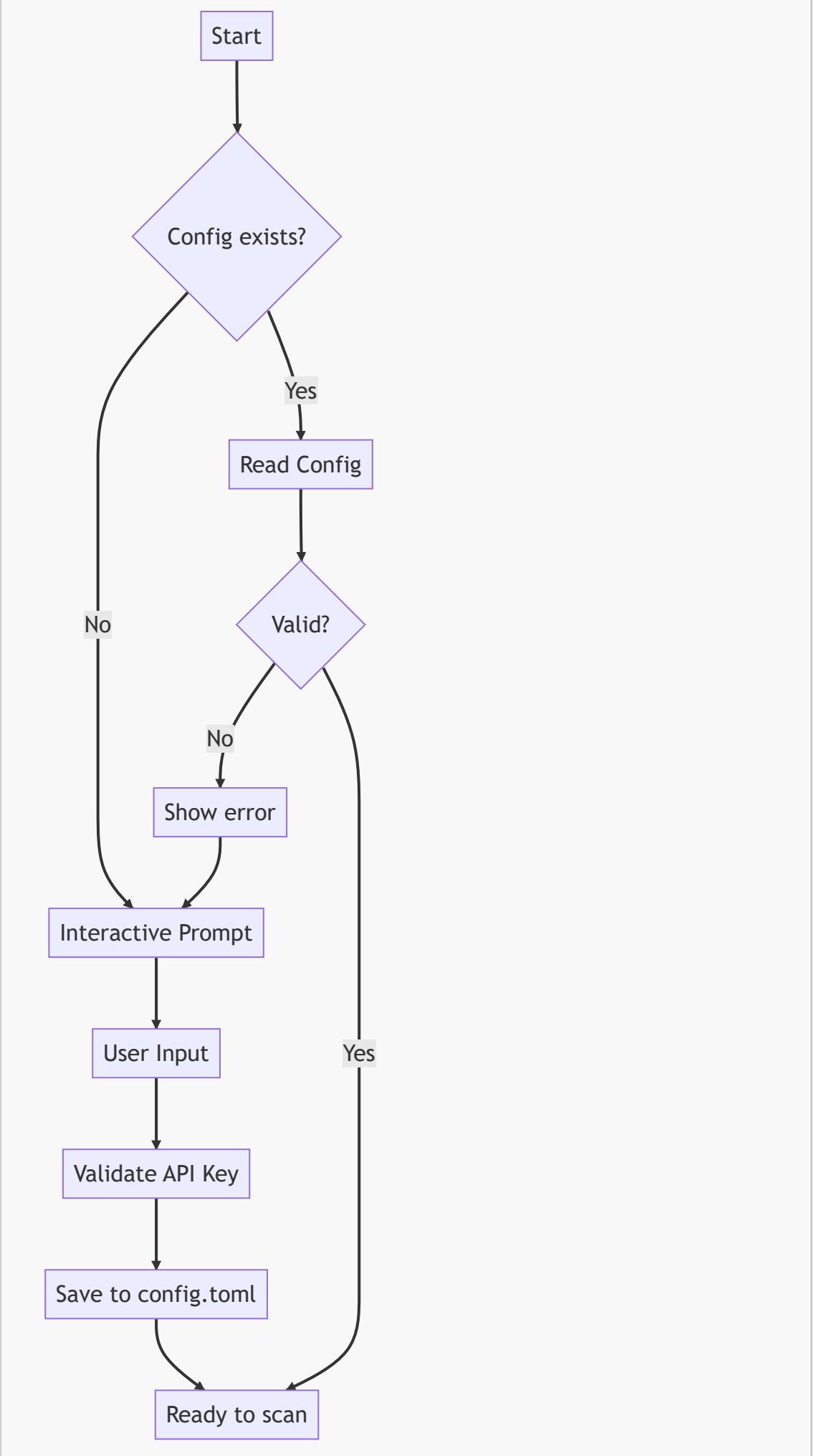
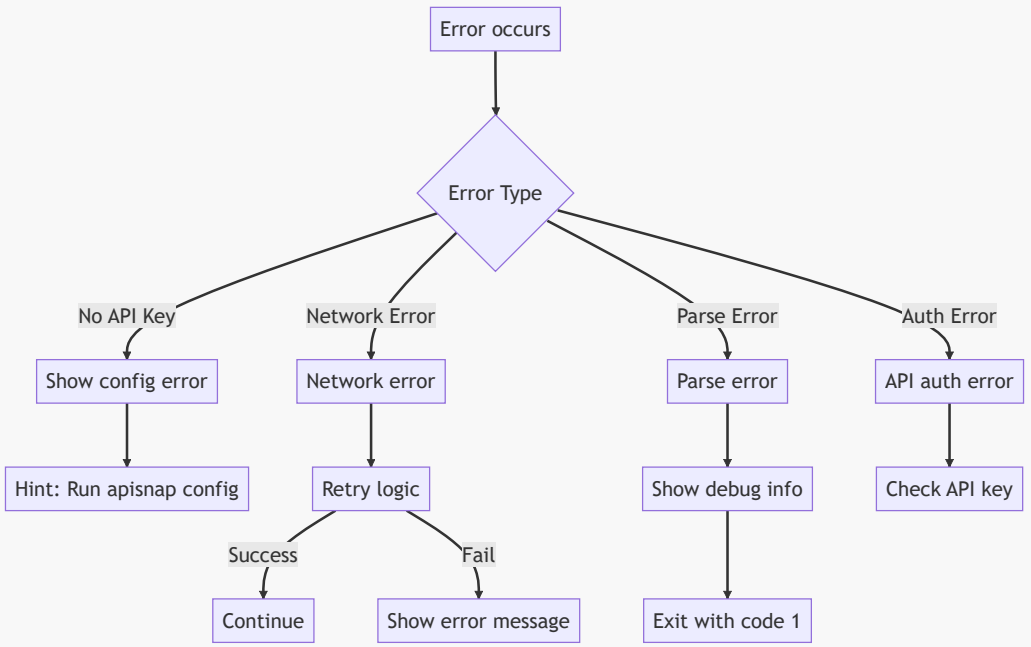

Start
Config exists?
Yes
No
Read Config
Valid?
No
Yes
Show error
Interactive Prompt
User Input
Validate API Key
Save to config.toml
Ready to scan

Diagram 8: Error Handling Flow



Internal Implementation

Core Data Structures

RouteManifest (schema.py)

```
@dataclass
class RouteManifest:
    """Represents a collection of API routes."""

    routes: list[Route] = field(default_factory=list)
    base_url: Optional[str] = None
    framework: Optional[str] = None
    project_name: Optional[str] = None
    source_mode: str = "unknown"
    detected_at: str = ""

    def to_dict(self) -> dict[str, Any]:
        return {
            "routes": [r.to_dict() for r in self.routes],
            "base_url": self.base_url,
            "framework": self.framework,
            "project_name": self.project_name,
            "source_mode": self.source_mode,
```

```
            "detected_at": self.detected_at,
        }
```

Route (schema.py)

```
@dataclass
class Route:
    """Represents an API route."""

    method: str  # GET, POST, PUT, DELETE, PATCH
    path: str    # /api/v1/users/{id}
    params: list[Param] = field(default_factory=list)
    body_schema: dict = field(default_factory=dict)
    response_schema: dict = field(default_factory=dict)
    auth_required: bool = False
    auth_type: Optional[str] = None  # "bearer", "api_key", "basic"
    tags: list[str] = field(default_factory=list)
    summary: Optional[str] = None
    source: str = "unknown"  # "openapi", "inferred", "github-data-
repo"
    confidence: float = 1.0  # 0.0 to 1.0
    refresh_schedule: Optional[str] = None  # cron expression
    public_url: Optional[str] = None  # full public URL
```

Param (schema.py)

```
@dataclass
class Param:
    """Represents an API parameter."""

    name: str
    location: str  # "path", "query", "header", "body"
    type: str       # "string", "integer", "boolean", "object",
"array"
    required: bool
    description: Optional[str] = None
    example: Optional[str] = None
```

GitHub Repository Scanner (github_repo_scanner.py)

```
class GitHubRepoScanner(BaseScanner):
    """Scanner for GitHub repos that use GitHub as a database."""

    GITHUB_API_BASE = "https://api.github.com"
    GITHUB_RAW_BASE = "https://raw.githubusercontent.com"
```

```
    JSON_FILE_PATTERNS = [
        "data/", "public/", "api/", "dist/",
        "output/", "assets/", "_data/", "json/", ""
    ]

    GENERATOR_PATTERNS = [
        ".github/workflows/", "scripts/",
        "fetch_", "update_", "sync_"
    ]

    def scan(self, url: str, **kwargs) -> RouteManifest:
        """Scan GitHub repository for JSON data files."""
        # 1. Parse owner/repo
        owner, repo = self._parse_github_url(url)

        # 2. Fetch repo tree
        tree = self._fetch_repo_tree(owner, repo)

        # 3. Parse workflows
        workflow_info = self._parse_workflows(owner, repo, tree)

        # 4. Find JSON files
        json_files = self._find_json_data_files(tree)

        # 5. Detect public URL
        base_url = self._detect_public_url(owner, repo, tree)

        # 6. Build routes
        routes = []
        for json_path in json_files:
            route = self._build_route(json_path, content, base_url)
            routes.append(route)

        return RouteManifest(routes=routes, base_url=base_url, ...)
```

AI Client (client.py)

```
class CerebrasClient:
    """Client for Cerebras AI API."""

    CEREBRAS_BASE_URL = "https://api.cerebras.ai/v1"
    DEFAULT_MODEL = "qwen-3-235b-a22b-instruct-2507"

    def generate_tests(self, route: Route, framework: str) -> str:
        """Generate tests for a route using Cerebras AI."""
        prompt = get_generate_tests_prompt(route, framework)

        response = self.client.chat.completions.create(
            model=self.model,
            messages=[
                {"role": "system", "content": "You are an API test
```

```
engineer."},
                {"role": "user", "content": prompt},
            ],
            temperature=0.2,
            max_tokens=4000,
        )

        return response.choices[0].message.content
```

Test Generation Prompts (prompts.py)

```
def get_generate_tests_prompt(route: Route, framework: str) -> str:
    """Generate comprehensive test prompt."""

    prompt = f"""Generate complete, runnable test code in
{framework}
    for the following API endpoint.

    Endpoint details:
    - Method: {route.method}
    - Path: {route.path}
    - Public URL: {route.public_url}
    - Auth required: {route.auth_required}
    - Response schema: {route.response_schema}

    Generate tests for ALL categories:
    1. Happy path (200 response)
    2. Schema field presence
    3. Field type validation
    4. Auth failure tests
    5. Wrong HTTP method (405)
    6. Empty/null response
    7. Content-Type check
    8. CORS headers
    9. Cache headers
    10. Boundary inputs
    11. 404 variations

    Rules:
    - Write complete, runnable code
    - No placeholders, no pseudocode
    - Each test needs clear name"""

    return prompt
```

Test Frameworks Supported

Python Frameworks

Framework	Command	Install	Use Case
pytest	`--format pytest`	`pip install requests pytest`	API testing
unittest	`--format unittest`	Built-in	Built-in testing
httpx	`--format httpx`	`pip install httpx pytest-httpx`	Async testing

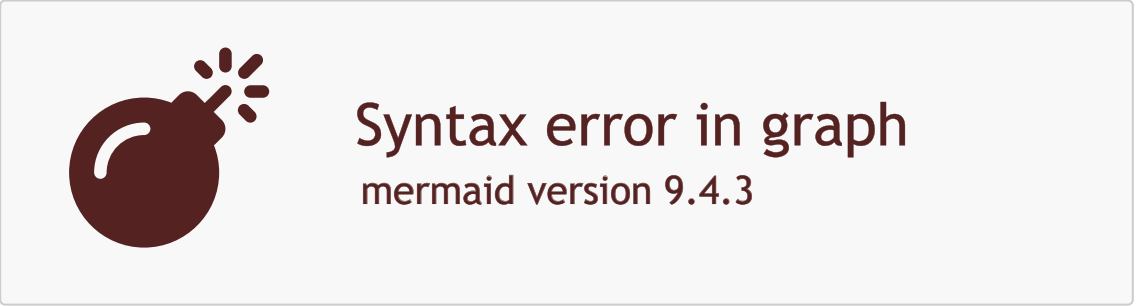
JavaScript Frameworks

Framework	Command	Install	Use Case
jest	`--format jest`	`npm install jest axios`	Unit/Integration
mocha	`--format mocha`	`npm install mocha axios`	BDD testing
vitest	`--format vitest`	`npm install vitest axios`	Vite-compatible

Other Frameworks

Framework	Command	Language
restassured	`--format restassured`	Java
rspec	`--format rspec`	Ruby

Framework Detection

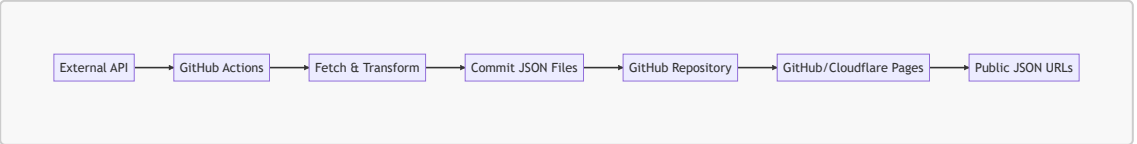


GitHub-as-Database Pattern

What is It?

The GitHub-as-Database pattern creates free, serverless JSON APIs using GitHub repositories for periodic data updates.

How It Works



Use Cases

- Weather data updates
- Stock/crypto prices
- Sports scores
- Currency exchange rates
- IoT sensor data
- Periodic reports

Example Workflow

```yaml
name: Update Data

on:
  schedule:
    - cron: '0 */6 * * *'  # Every 6 hours

jobs:
  update:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - name: Fetch data
        run: python fetch_prices.py
      - name: Commit
        uses: stefanzweifel/git-auto-commit-action@v5
        with:
          commit_message: "Update prices"
          file_pattern: "data/*.json"
```

What apisnap Detects

1. **Workflow schedules** - cron expressions
2. **Data source URLs** - external API calls
3. **Output file patterns** - data/*.json
4. **Serving URLs** - GitHub Pages, Cloudflare Pages
5. **Data schemas** - JSON structure analysis

Best Practices

1. API Key Management

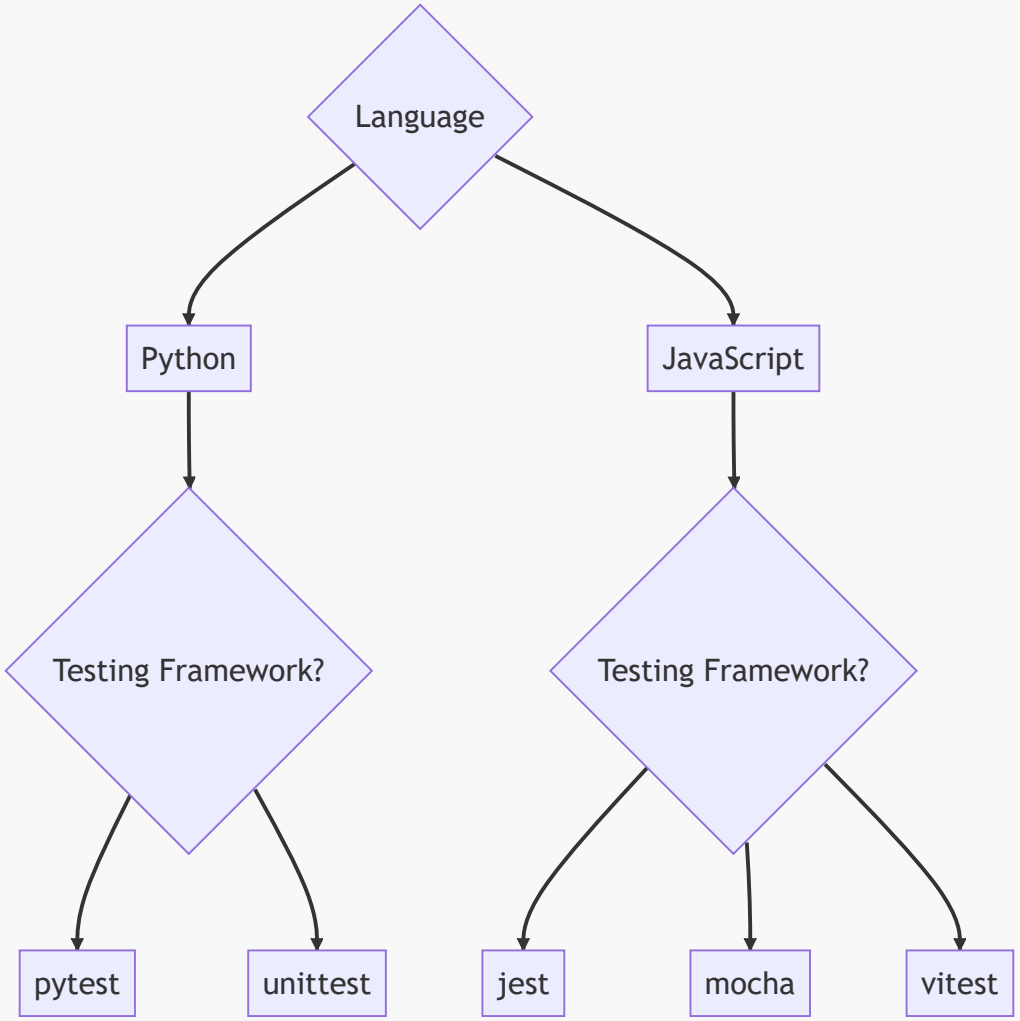
```bash
# Set API key (non-interactive)
apisnap config --api-key $CEREBRAS_API_KEY

# Or use environment variable
```

```
export CEREBRAS_API_KEY="csk-xxxx"
apisnap config --api-key $CEREBRAS_API_KEY
```

2. Output Organization

```
# Use separate test directories per project
apisnap scan --url REPO --output ./tests/REPO
```

3. Framework Selection



4. CI/CD Integration

```
# .github/workflows/test-generation.yml
name: Generate API Tests

on:
  push:
    branches: [main]
```

```yaml
  schedule:
    - cron: '0 0 * * *'  # Daily

jobs:
  generate-tests:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: astral-sh/uv/setup-python@v1

      - name: Set API key
        run: apisnap config --api-key ${{ secrets.CEREBRAS_API_KEY
}}

      - name: Generate tests
        run: apisnap scan --url ${{ github.repository }} --format
pytest

      - name: Run tests
        run: pytest -v

      - name: Commit tests
        if: always()
        uses: stefanzweifel/git-auto-commit-action@v5
        with:
          commit_message: "Update API tests"
```

5. Handling Errors

```bash
# Enable verbose for debugging
apisnap scan --url REPO --verbose

# Check routes without generating
apisnap scan --url REPO --dry-run

# Skip AI for quick preview
apisnap scan --url REPO --no-ai
```

Troubleshooting

Common Issues

Issue	Solution
No API key	Run `apisnap config --api-key <key>`
Network error	Check connection, try verbose mode
No routes found	Verify the repo has JSON files in data/, public/

Issue	Solution
Low confidence	Manually review generated tests
Wrong framework	Use `--framework` flag to force

Error Messages

```
# No API key
Error: No API key configured. Run 'apisnap config' first.

# Network error
Error: Failed to scan: Connection refused

# Parse error
Error: Failed to parse OpenAPI spec
```

Integration Guide

Integration with Existing Projects

```
# 1. Add apisnap to project
uv add apisnap --dev

# 2. Generate initial tests
apisnap scan ./src --output ./tests/generated

# 3. Review generated tests
ls tests/generated/

# 4. Run tests
pytest tests/generated/ -v

# 5. Add to version control
git add tests/generated/
git commit -m "Add generated API tests"
```

GitHub Actions Integration

```
name: API Tests

on:
  push:
    paths:
      - 'src/**'
  workflow_dispatch:
```

```yaml
jobs:
  api-tests:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Generate tests
        run: |
          apisnap config --api-key ${{ secrets.CEREBRAS_API_KEY }}
          apisnap scan --url ${{ github.repository }} --format
pytest

      - name: Run tests
        run: pytest -v --junitxml=report.xml

      - name: Upload results
        uses: actions/upload-artifact@v4
        if: always()
        with:
          name: test-results
          path: report.xml
```

Custom Test Framework Integration

To add a new test framework:

1. Create writer in `src/apisnap/writers/<framework>_writer.py`

```python
from apisnap.writers.base_writer import BaseWriter

class CustomWriter(BaseWriter):
    def write(self, manifest: RouteManifest, output: str) ->
list[str]:
        """Generate tests in custom framework."""
        # Implementation
        pass
```

2. Register in `src/apisnap/writers/__init__.py`

CLI Command Reference

Quick Reference

```
# Configuration
apisnap config --api-key <key>          # Set API key
apisnap config --show                   # Show config
apisnap config --format pytest          # Set default format
```

```
# Scanning
apisnap scan <path>                    # Scan local
apisnap scan --url <url>                # Scan remote
apisnap scan --dry-run                  # Preview only
apisnap scan --no-ai                    # Skip AI

# Output
apisnap scan --format <framework>        # Specify format
apisnap scan --output <dir>            # Output directory
apisnap scan --verbose                 # Verbose

# Other
apisnap list                           # List routes
apisnap version                       # Show version
```

Additional Resources

- PyPI Package
- Cerebras AI
- GitHub Actions Documentation
- Cloudflare Pages

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